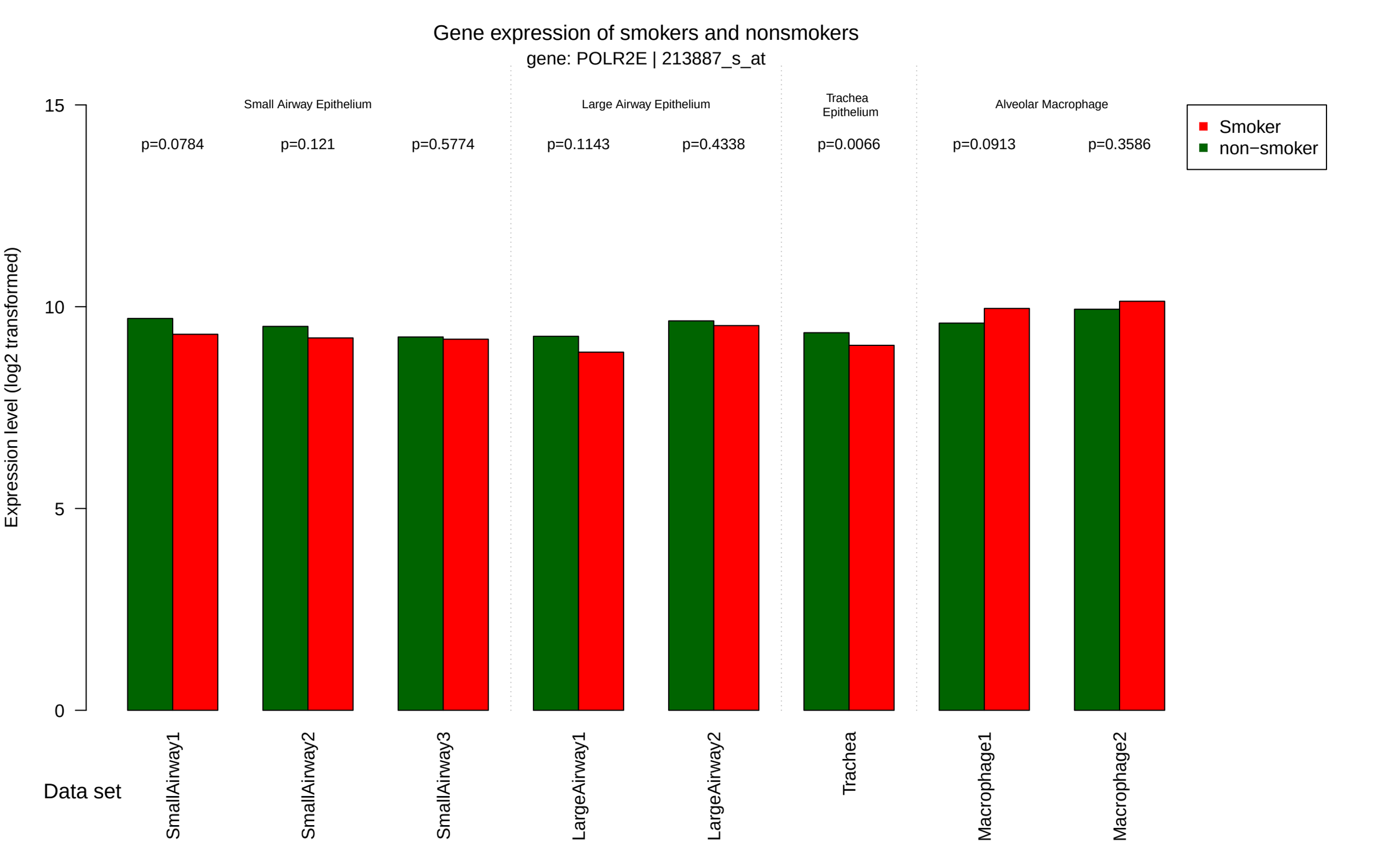
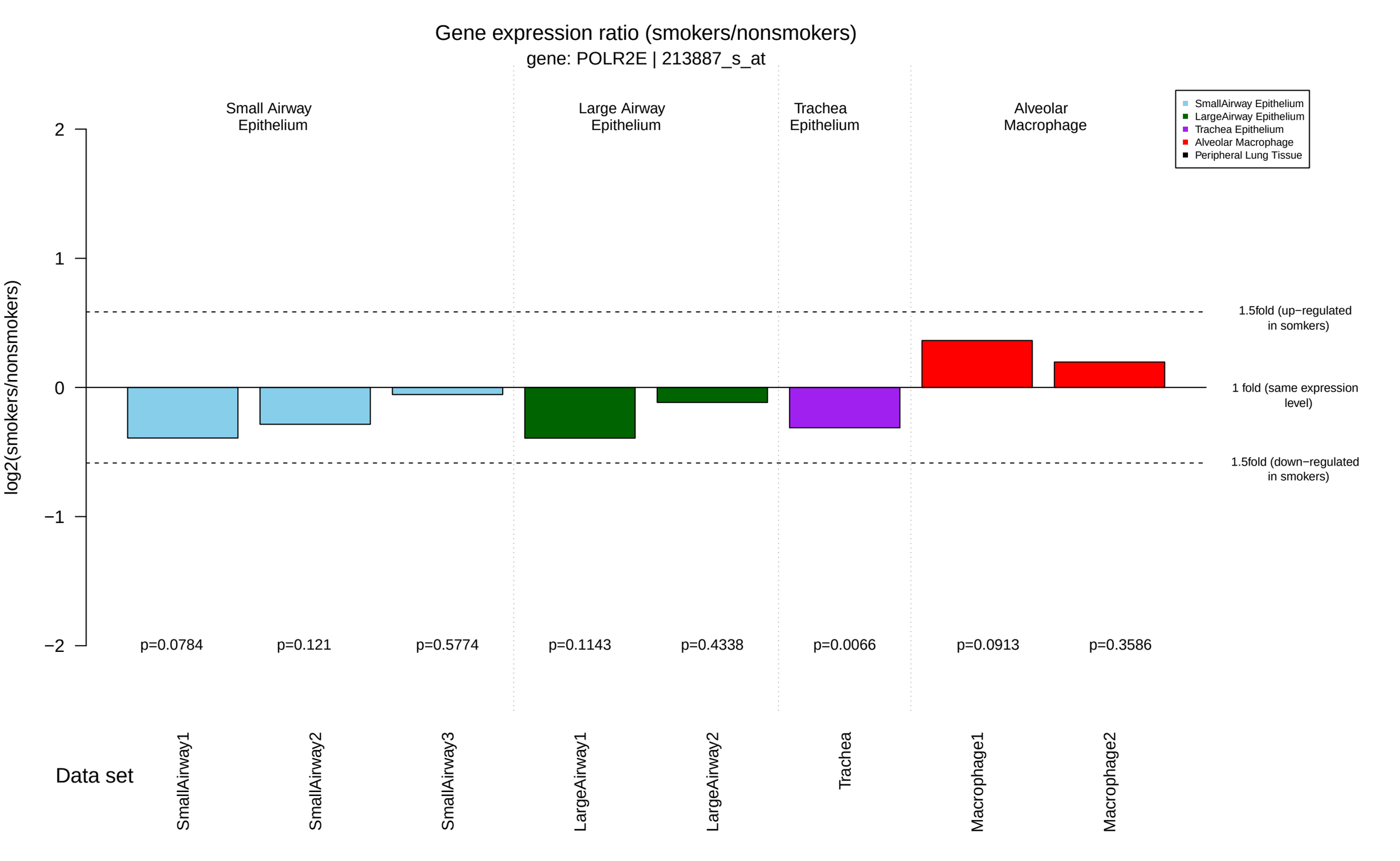
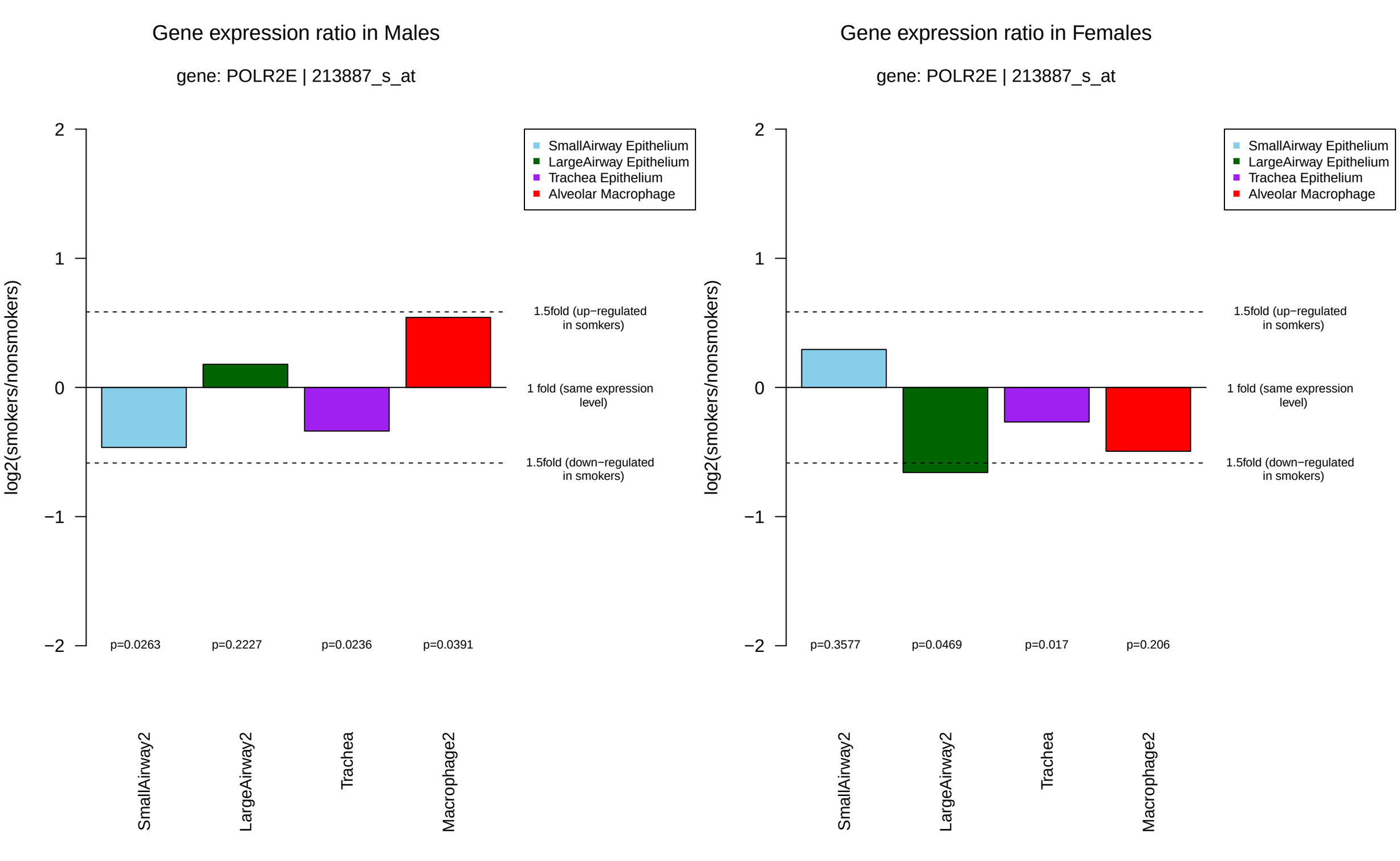
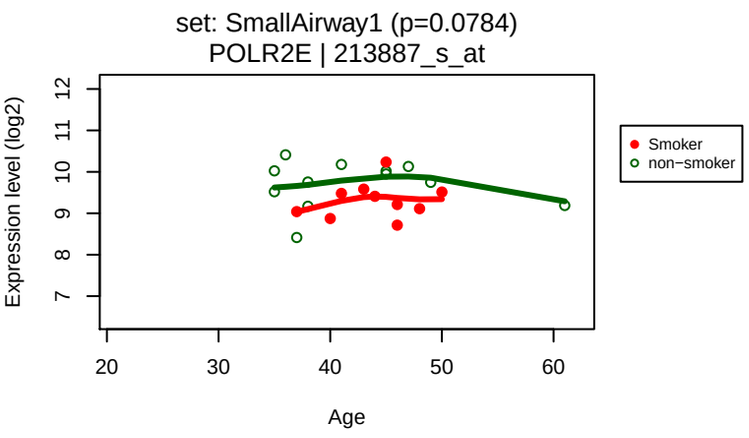
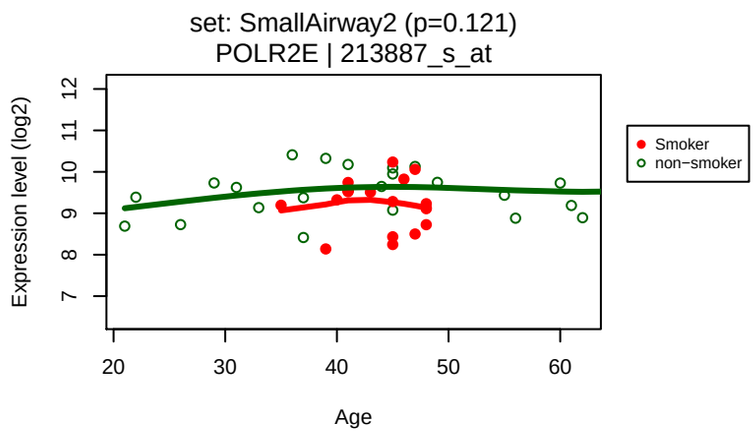
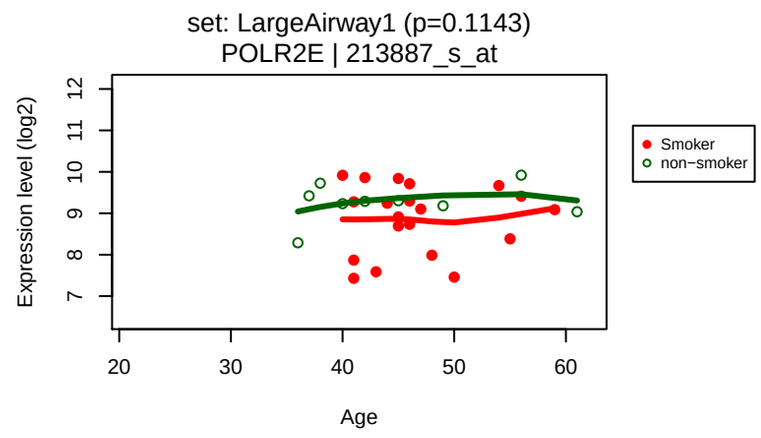

Gene expression of smokers and nonsmokers
gene: POLR2E | 213887_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0784
p=0.121
p=0.5774
p=0.1143
p=0.4338
p=0.0066
p=0.0913
p=0.3586
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

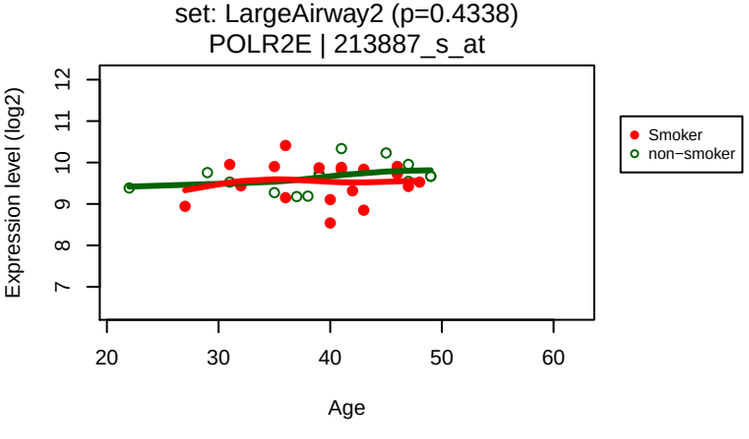

Gene expression ratio (smokers/nonsmokers)
gene: POLR2E | 213887_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0784
p=0.121
p=0.5774
p=0.1143
p=0.4338
p=0.0066
p=0.0913
p=0.3586
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

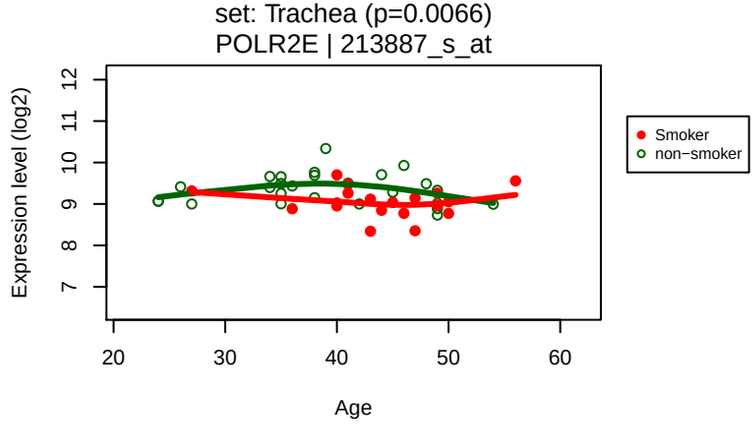

Gene expression ratio in Males
gene: POLR2E | 213887_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0263
p=0.2227
p=0.0236
p=0.0391
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: POLR2E | 213887_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3577
p=0.0469
p=0.017
p=0.206
SmallAirway2
LargeAirway2
Trachea
Macrophage2

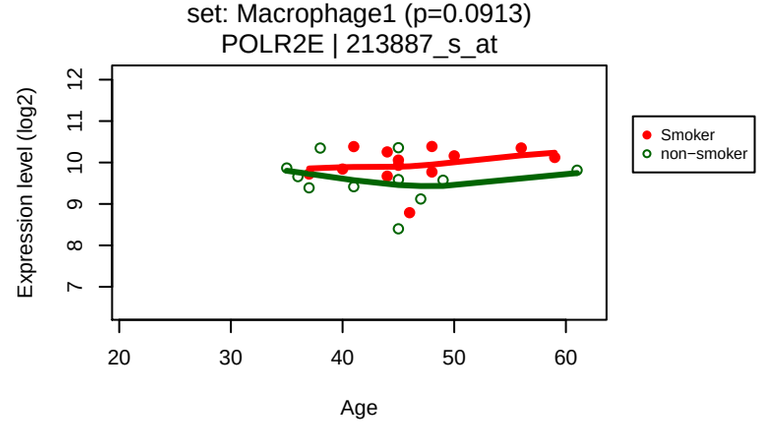

set: SmallAirway1 (p=0.0784)
POLR2E | 213887_s_at
Expression level (log2)
Age
Smoker
non−smoker

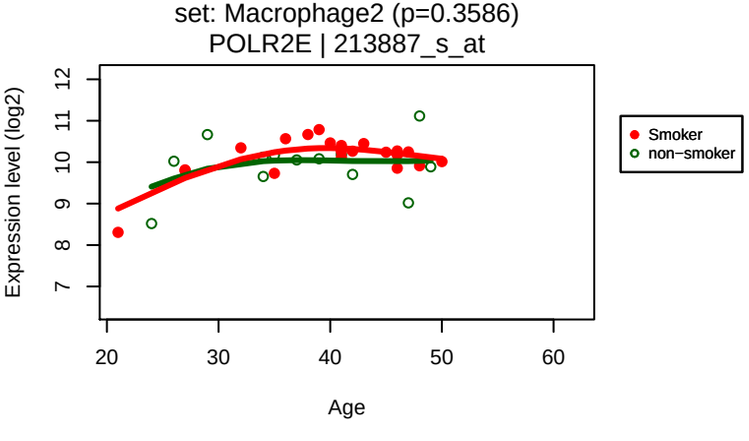

set: SmallAirway2 (p=0.121)
POLR2E | 213887_s_at
Expression level (log2)
Age
Smoker
non−smoker

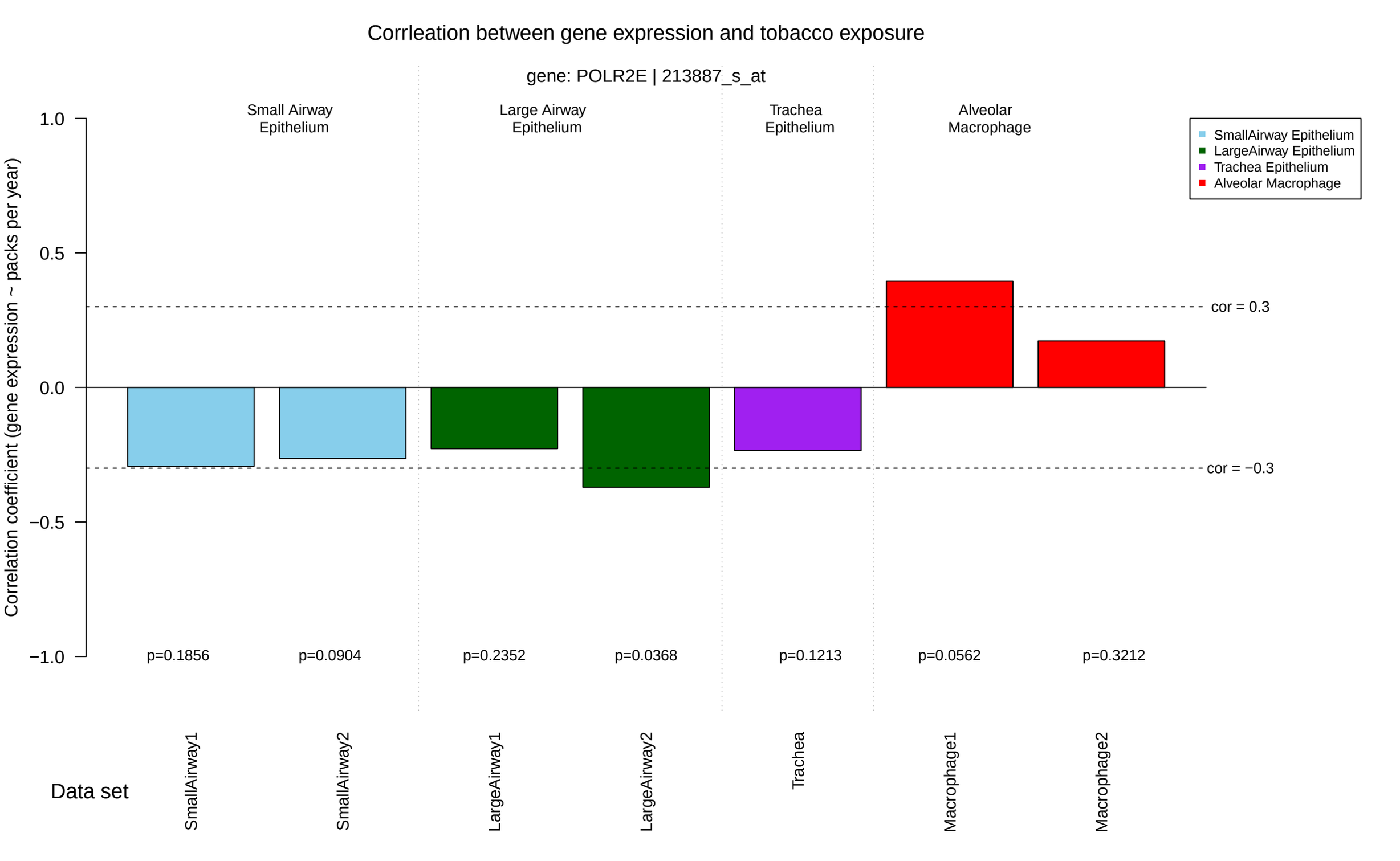

set: LargeAirway1 (p=0.1143)
POLR2E | 213887_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4338)
POLR2E | 213887_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0066)
POLR2E | 213887_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0913)
POLR2E | 213887_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3586)
POLR2E | 213887_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: POLR2E | 213887_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1856
p=0.0904
p=0.2352
p=0.0368
p=0.1213
p=0.0562
p=0.3212
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2